Title page

Title. EIG-MPHBS: a rotation-invariant information exchange for hybrid Honey Badger–Marine Predators optimisation

Author. Abinaya G[a,*]

[a]Department of Information Technology, Saveetha Engineering College, Saveetha Nagar, Thandalam, Chennai 602105, Tamil Nadu, India

[*]**Corresponding author.** Abinaya G, Department of Information Technology, Saveetha Engineering College, Saveetha Nagar, Thandalam, Chennai 602105, Tamil Nadu, India. E-mail: `gabinaya@saveetha.ac.in`

Acknowledgements. The author thanks the authors of MPHBS for publishing their code and per-run results, and the organisers of the CEC 2022 competition on single objective bound constrained numerical optimisation for publishing the competitors' result files.

Declaration of competing interest. The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement. Abinaya G: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Data and code availability. Code, the design document, the record of every run, the analyses and the scripts that generate every number and table are available at `https://github.com/abinaya-g/la-kaie` (code on branch `eswa/eigen-exchange`, run records on branch `eswa/results`). The per-run convergence and population traces used for the figures (1.6 GB) are available from the author on request. This link is withheld from the anonymized manuscript.

Reference withheld for anonymized review. The anonymized manuscript cites "Anonymous (2026)". It is the author's manuscript "Does the learning matter? A pre-registered deconstruction of SARSA-guided information exchange in the MPHBS hybrid metaheuristic", which is under review elsewhere. It shows that the SARSA learner of MPHBS gives no benefit over a random choice at the same budget; the present manuscript uses that random-mediator variant as a baseline, in new runs, and its contribution (the eigen-frame exchange) does not appear in that manuscript.